

The waveform-first forward-fit reconstruction for the MX17 micro-TPC

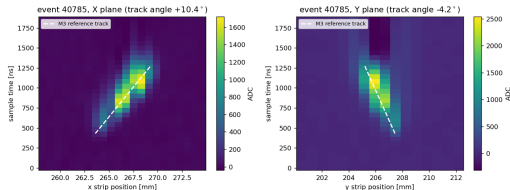
threading solved · v_{drift} resolved · gas identified · in-situ gap maps

MX17 June cosmic-bench analysis

25–27 July 2026

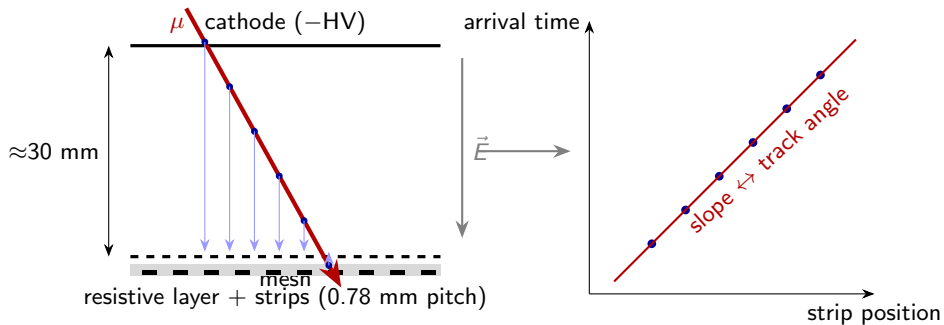
The starting problem: tracks don't “thread” the reference

- Micro-TPC: strip position + arrival time $\Rightarrow$ 3-D track per muon
- Hit-level reconstruction: fitted ladders $\sim 4^\circ$ **too steep**; the M3 reference threads the cloud at the mesh, fans away with depth
- Not alignment ($< 10\ \mu\text{m}$); “unsharing” fixed it only on average
- **This work: rebuild from raw waveforms, zero assumptions**



one muon in the raw waveform matrix; dashed = M3 prediction

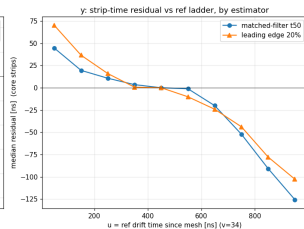
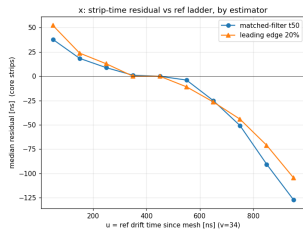
How the micro-TPC works



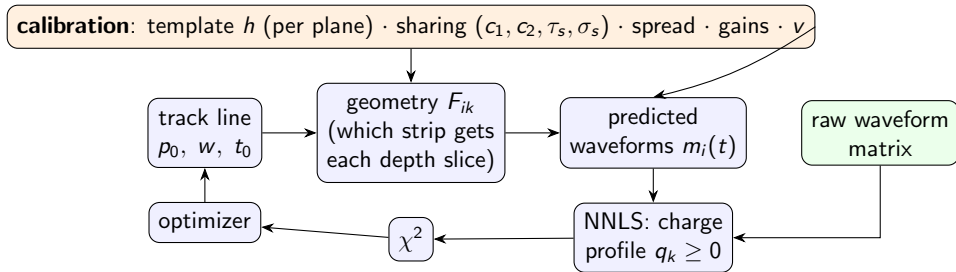
Arrival time encodes depth. A straight muon = a straight line in (position, time).

Why every hit-based timing fails: the sharing mixture

- The resistive layer sends each strip's signal to its neighbours: **delayed** ($\tau_s \approx 47$ ns) and **dispersed** ($\sigma_s \approx 90$ ns), $c_1 \approx 29\%$ each side
- Neighbours' charge arrived at *different depths/times*
- Any time extracted from the aggregate is pulled toward the cluster mean
- Measured: +40 ns late at the mesh, -100 ns early at the cathode, **same for every estimator** $\Rightarrow$ ladder compressed 20–30% $\Rightarrow$ “4° too steep”



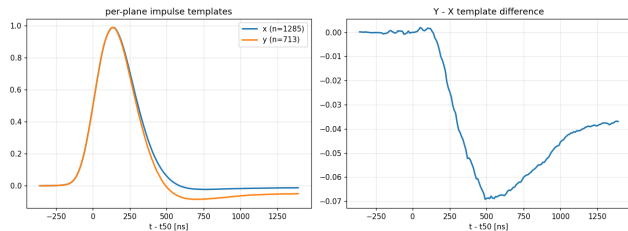
The fix: fit the waveforms, not the hits



$$m_i(t) = \sum_k q_k [F_{ik} h(t - t_0 - u_k) + c_1(F_{i\pm 1, k}) h_s(t - t_0 - u_k - \tau_s) + \dots]$$

- Sharing modelled **forward** (well-conditioned) — never deconvolved
- Linear in the charges (fast NNLS); 3 nonlinear track parameters; ~ 0.1 s/plane
- $w = v \tan \theta$ in mm/ns: **no drift velocity assumed in the fit**

Ingredients are measured, per detector



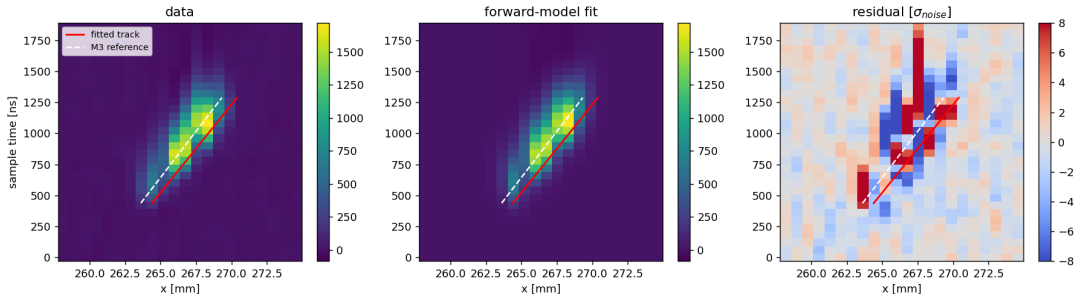
impulse template from bright steep-track strips; Y undershoot $4\times$ deeper

Calibration protocol (automated):

- 1 template per plane (data-driven)
- 2 gain map (det3: 1.4% spread only)
- 3 8-parameter hyper-fit with track pinned to M3 on ~ 150 muons
($c_1, c_2, k_Y, \tau_s, \sigma_s, \sigma_{p0}, D_p, v$)

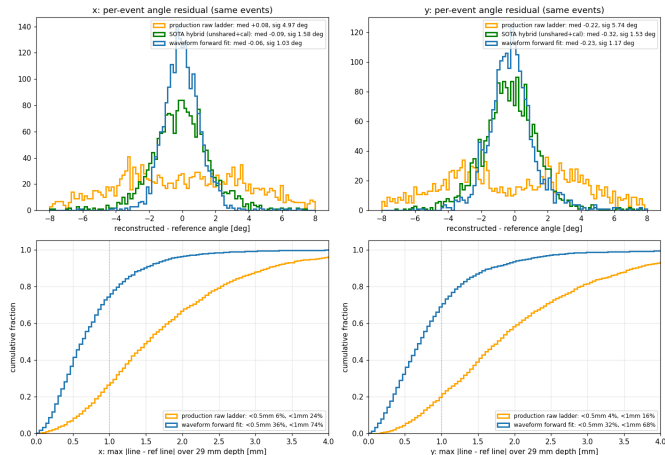
Anti-circularity: calibrating against the biased ladders instead is 49% worse in χ^2 — the waveforms *prefer* the reference tracks.

One event, fitted



data · model · residual; fitted 10.9° vs M3 10.4°

Performance: head-to-head on identical events



	angle σ	<1mm depth
raw ladder	5.0/5.7°	24/16%
SOTA hybrid	1.6/1.5°	—
forward fit	1.06/1.10°	74/70%

- mesh position $\sigma = 0.57/0.59$ mm
- unbiased ($< 0.3^\circ$), flat implied- v vs angle (physical)
- all angles: $\sim 1.0\text{--}1.1^\circ$ for $|\tan \theta| > 0.04$

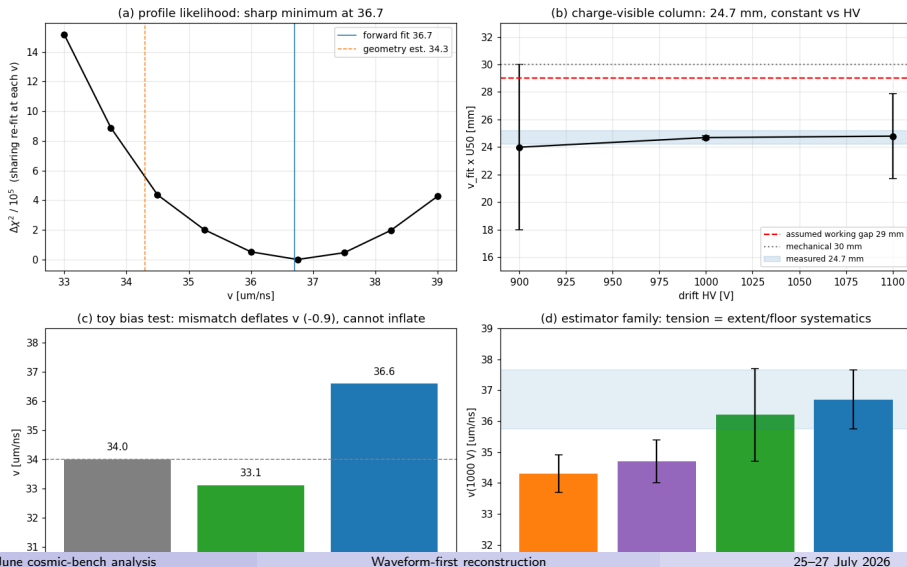
The remaining 1° is physics, not reconstruction

toy truth	angle σ recovered	$p_0 \sigma$
electronics noise only	0.02°	$7 \mu\text{m}$
+0.15 mm/bin charge-centroid jitter	0.55°	$219 \mu\text{m}$
+0.30 mm/bin (diffusion scale)	1.05°	$354 \mu\text{m}$
+0.50 mm/bin	1.77°	$473 \mu\text{m}$

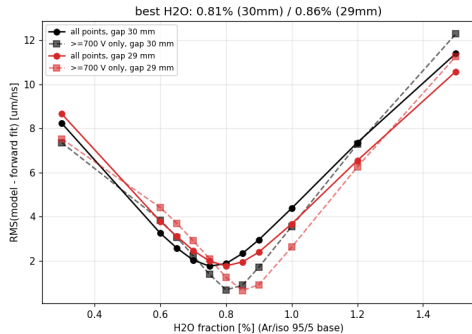
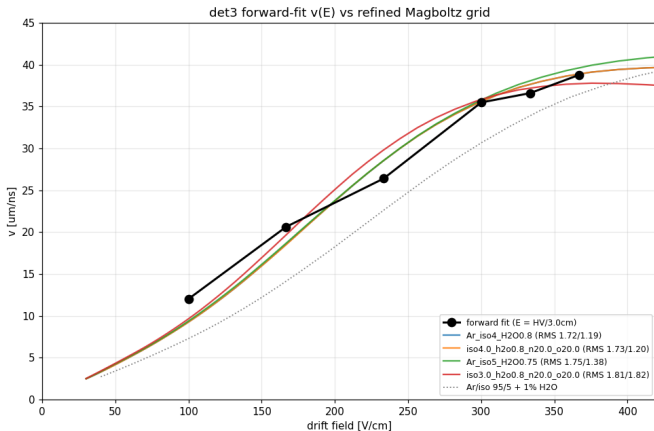
- Toy closure: model-generated waveforms with known tracks, real noise/geometry
- The observed $1.05^\circ = \text{stochastic diffusion} + \text{avalanche-weighted charge granularity}$ — the detector's information limit
- Ablations: no single model component moves the width by $> 0.05^\circ \Rightarrow$ a production implementation can stay simple

The drift-velocity tension: resolved

det3 drift-velocity tension resolution (2026-07-26)



The gas, identified by Magboltz

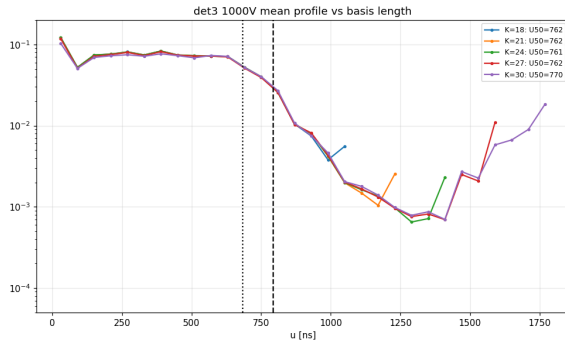


- 71 mixtures (41 new: local + condor; cross-validated to $0.01 \mu\text{m}/\text{ns}$)
- Ar/ $i\text{C}_4\text{H}_{10}$ 95/5 + 0.8% H_2O , RMS $0.63 \mu\text{m}/\text{ns}$ (≥ 700 V)**
- consistent with the week's bottle drying

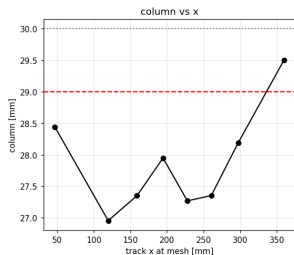
A scare that became a measurement: the drift-gap maps

The deconvolved charge profile's endpoint U measures the local gap: $g = v \cdot U$.

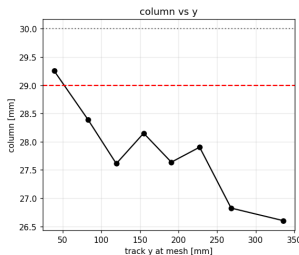
- first pass said 24.7 mm (!) — traced to two estimator biases (median over sparse NNLS; free-fit profiles)
- corrected: $U_{50} = 762 \text{ ns} \Rightarrow \mathbf{27.9 \text{ mm}}$, stable vs basis length, **closure-validated** (toys recover known columns to $\leq 5 \text{ ns}$, even with wrong-undershoot generation)
- the edge is *soft* where toy steps are sharp $\Rightarrow$ **the gap varies with position**



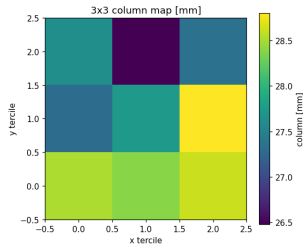
det3's cathode is tilted; det4 bowed; det2 flat



det2 (B): flat, 30.5 mm



det3 (A): tilted ~ 2.7 mm



det4 (E): bowed ± 3 mm



mild E^2 electrostatic pull on top (28.4 $\rightarrow$ 27.8 mm, 900 $\rightarrow$ 1100 V). **No short gap; no disassembly required** —
 $U_{50}(x, y)$ is a free in-situ flatness survey per chamber.

The production stack, validated on three chambers

	det3 (A)	det2 (B)	det4 (E)
kernel $c_1/k_Y/\tau_s$	0.29/1.38/47 ns	0.29/1.42/49 ns	0.24/2.36/68 ns
v (run conditions)	36.6 $\mu\text{m}/\text{ns}$	39.9 $\mu\text{m}/\text{ns}$	34.2 $\mu\text{m}/\text{ns}$
angle σ (x/y)	1.06/1.10 $^\circ$	1.33/1.54 $^\circ$	2.00/1.97 $^\circ$
mesh σ (x/y)	0.57/0.59 mm	0.76/0.56 mm	0.93/0.71 mm
column / topography	27.9 mm, y-tilt	30.5 mm, flat	28.8 mm, bowed

- `wft_reco.py: fit_event()` $\rightarrow$ position, angle, charge profile, honest errors (stat $\oplus$ physics floor), quality flags, near-vertical fallback
- calibration scripts fully automated; handles different DAQ windows
- plus: $\chi^2(v)$ gas monitor, gap mapper, cheap monitoring pre-pass

Conclusions

- **Threading solved:** the divergence was aggregate-time mixing from resistive sharing; the forward fit removes it per event
- $\sigma \approx 1^\circ$ / **0.6 mm per event** — **at the physics floor** (diffusion + charge granularity; noise contributes 0.02°)
- $v_{\text{drift}} = 36.7 \pm 0.3 \pm 0.9 \mu\text{m}/\text{ns}$; gas = wet 95/5 (0.8% H₂O), Magboltz-matched across the full drift scan
- **Drift-gap topography measured in situ:** det3 tilted 2.7 mm, det4 bowed, det2 flat at 30.5 mm
- **Next:** mechanical flatness check at next opening; humidity probe; 300/500 V re-scan; det6/7; numba port; July beam data

Report: waveform_first_threading/report/main.pdf · full trail: WAVEFORM_FIRST_THREADING.md · scripts 01–36